

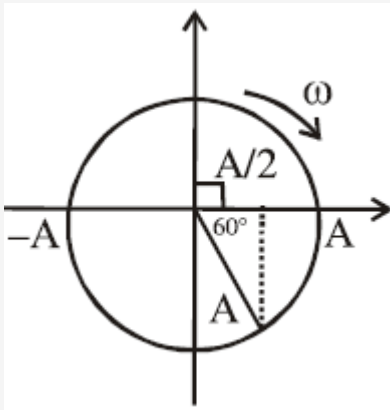
Q1: $Y = A \sin(\omega t + \phi_0)$ is the time-displacement equation of a SHM. At $t = 0$ the displacement of the particle is $Y = \frac{A}{2}$ and it is moving along negative x-direction. Then the initial phase angle ϕ_0 will be:

- (A) $\frac{\pi}{6}$
- (B) $\frac{5\pi}{6}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{2\pi}{3}$

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- (D) $\frac{2\pi}{3}$

Solution:



Initial phase $\frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$

Q2: For extrinsic semiconductors; when doping level is increased;

- (A) Fermi-level of p and n-type semiconductors will not be affected.
- (B) Fermi-level of p-type semiconductors will go downward and Fermi-level of n-type semiconductor will go upward.
- (C) Fermi-level of p-type semiconductor will go upward and Fermi-level of n-type semiconductors will go downward.
- (D) Fermi-level of both p-type and n-type semiconductors will go upward for $T > T_F$ K and downward for $T < T_F$ K, where T_F is Fermi temperature.

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Solution:

In an extrinsic semiconductor (with added doping), in order to conserve the number of particles and to fulfill the overall electrical charge neutrality, the Fermi level has to move away from the midgap position.

It has to shift towards conduction band in an n-type semiconductor (extrinsic semiconductors with added doping impurities which are donors i.e. impurities which give additional electrons to the system) where the number of electrons n is higher than the number of holes ($n > p$).

It shifts towards valence band in a p-type semiconductor (extrinsic semiconductors with added doping impurities which are acceptors i.e. impurities which trap electrons from the system giving rise to a deficit of electrons or an excess of holes) where the number of electrons n is lower than the number of holes ($n < p$).

Q3: Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter $0.1 \mu m$. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern such that:

- (A) Its size decreases, but intensity increases
- (B) Its size increases, and intensity increases
- (C) Its size increases, but intensity decreases
- (D) Its size decreases, and intensity decreases

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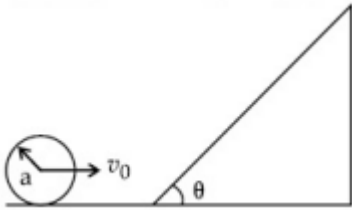
Solution:

$$\sin \theta = \frac{m\lambda}{a}$$

When a increases, θ decreases, width decreases.

If width decreases then intensity increases.

Q4: A sphere of radius 'a' and mass 'm' rolls along a horizontal plane with constant speed v_0 . It encounters an inclined plane at angle θ and climbs upward. Assuming that it rolls without slipping, how far up the sphere will travel?



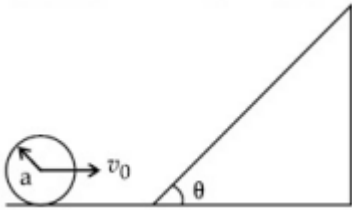
(A) $\frac{5v_0^2 \sin \theta}{g}$

(B) $\frac{7v_0^2 \sin \theta}{10g}$

(C) $\frac{5}{2} \frac{v_0^2 \sin \theta}{g}$

(D) $\frac{2v_0^2 \sin \theta}{g}$

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(D) $\frac{2v_0^2 \sin \theta}{g}$

Solution:

By energy conservation

$$\frac{1}{2}mv_o^2 + \frac{1}{2}I\omega^2 = mgh$$

$$\frac{1}{2}mv_o^2 + \frac{1}{2} \times \frac{2}{5}mr^2 \times \left(\frac{v_o}{r}\right)^2 = \frac{mgl}{\sin \theta}$$

$$\frac{7}{10}mv_o^2 = \frac{mgl}{\sin \theta}$$

$$l = \frac{7v_o^2 \sin \theta}{10g}$$

Q5: Given below are two statements:

Statement I: In a diatomic molecule, the rotational energy at a given temperature obeys Maxwell's distribution.

Statement II: In a diatomic molecule, the rotational energy at a given temperature equals the translational kinetic energy for each molecule.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both Statement I and Statement II are true.
- (B) Statement I is false but Statement II is true.
- (C) Both Statement I and Statement II are false.
- (D) Statement I is true but Statement II is false.

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Solution:

Translational degree of freedom = 3

Rotational degree of freedom = 2

The translational kinetic energy & rotational kinetic energy both obey Maxwell's distribution independent of each other.

So, second statement is false.

Q6: A stone is dropped from the top of a building. When it crosses a point 5 m below the top, another stone starts to fall from a point 25 m below the top. Both stones reach the bottom of building simultaneously. The height of the building is:

- (A) 50 m
- (B) 35 m
- (C) 45 m
- (D) 25 m

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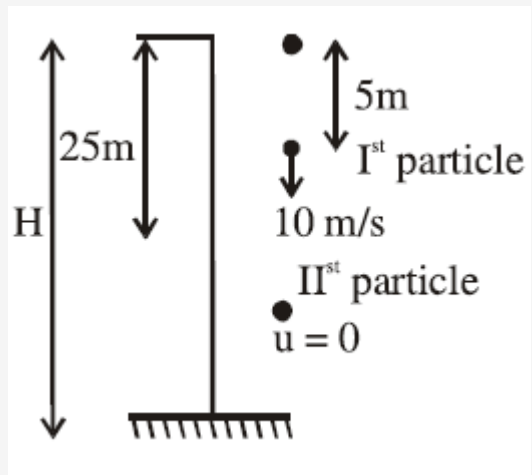
(A) 50 m

(B) 35 m

(C) 45 m

(D) 25 m

Solution:



$$\text{Time for particle to meet} = t' = \frac{S_{\text{rel}}}{v_{\text{rel}}} = \frac{20}{10} = 2 \text{ sec}$$

$$\text{Time taken by } I^{\text{st}} \text{ particle to reach ground} = 3 \text{ sec}$$

$$H = \frac{1}{2}g(3)^2 = 45\text{m}$$

Q7: In a ferromagnetic material, below the curie temperature, a domain is defined as:

- (A) a macroscopic region with saturation magnetization.
- (B) a macroscopic region with zero magnetization.
- (C) a macroscopic region with consecutive magnetic dipoles oriented in opposite direction.
- (D) a macroscopic region with randomly oriented magnetic dipoles.

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Solution:

In ferromagnetic materials, smaller groups of atoms band together into areas called domains, in which all the electrons have the same magnetic orientation.

Q8: Match List I with List II.

List I	List II
(a) Rectifier	(i) Used either for stepping up or stepping down the a.c. voltage
(b) Stabilizer	(ii) Used to convert a.c. voltage into d.c. voltage
(c) Transformer	(iii) Used to remove any ripple in the rectified output voltage
(d) Filter	(iv) Used for constant output voltage even when the input voltage or load current change

Choose the correct answer from the options given below:

- (A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- (B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
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- (C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (D) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Solution:

- (a) Rectifier → **AC** to DC
- (b) Stabilizer → used for constant output voltage even when voltage or current change.
- (c) Transformer → Step – up or Step – down ac voltage.
- (d) Filter → Used to remove any ripple in the rectified output voltage.

Q9: If a message signal of frequency ' f_m ' is amplitude modulated with a carrier signal of frequency ' f_c ' and radiated through an antenna, the wavelength of the corresponding signal in air is:

(A) $\frac{c}{f_c}$

(B) $\frac{c}{f_m}$

(C) $\frac{c}{f_c - f_m}$

(D) $\frac{c}{f_c + f_m}$

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(B) $\frac{c}{f_m}$

(C) $\frac{c}{f_c - f_m}$

(D) $\frac{c}{f_c + f_m}$

Solution:

$$\lambda = \frac{v}{f} = \frac{c}{f_c}$$

Q10: An electron of mass m_e and a proton of mass $m_p = 1836 m_e$ are moving with the same speed. The ratio of their de Broglie wavelength $\frac{\lambda_{electron}}{\lambda_{proton}}$ will be:

(A) 1836

(B) 918

(C) 1

(D) $\frac{1}{1836}$

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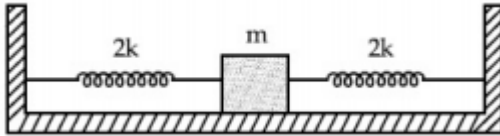
(C) 1

(D) $\frac{1}{1836}$

Solution:

$$\frac{\lambda_e}{\lambda_p} = \frac{\frac{h}{m_e v}}{\frac{h}{m_p v}} = 1836$$

Q11: Two identical springs of spring constant '2k' are attached to a block of mass m and to fixed support (see figure). When the mass is displaced from equilibrium position on either side, it executes simple harmonic motion. The time period of oscillations of this system is:



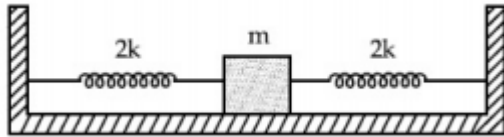
(A) $2\pi\sqrt{\frac{m}{2k}}$

(B) $2\pi\sqrt{\frac{m}{k}}$

(C) $\pi\sqrt{\frac{m}{k}}$

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(D) $\pi\sqrt{\frac{m}{2k}}$

Solution:

For parallel combination $k_{eq} = k_1 + k_2$

$$k_{eq} = 4k$$

$$T = 2\pi\sqrt{\frac{m}{k_{eq}}}$$

$$T = 2\pi\sqrt{\frac{m}{4k}}$$

$$T = \pi\sqrt{\frac{m}{k}}$$

Q12: An electron with kinetic energy K_1 enters between parallel plates of a capacitor at an angle ' α ' with the plates. It leaves the plates at angle ' β ' with kinetic energy K_2 . Then the ratio of kinetic energies $K_1 : K_2$ will be:

(A) $\frac{\sin^2 \beta}{\cos^2 \alpha}$

(B) $\frac{\cos \beta}{\sin \alpha}$

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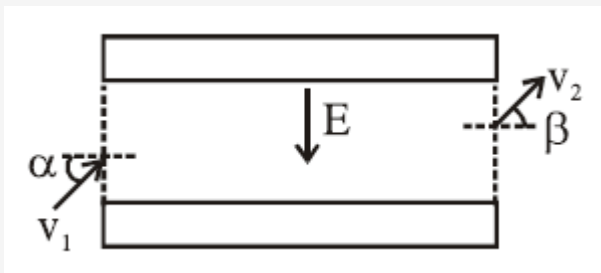
(A) $\frac{\sin^2 \beta}{\cos^2 \alpha}$

(B) $\frac{\cos \beta}{\sin \alpha}$

(C) $\frac{\cos \beta}{\cos \alpha}$

(D) $\frac{\cos^2 \beta}{\cos^2 \alpha}$

Solution:



Velocity along the plate will not change.

$$\therefore v_1 \cos \alpha = v_2 \cos \beta$$

$$\frac{K_1}{K_2} \Rightarrow \frac{v_1^2}{v_2^2} = \frac{\cos^2 \beta}{\cos^2 \alpha}$$

Q13: The wavelength of the photon emitted by a hydrogen atom when an electron makes a transition from $n = 2$ to $n = 1$ state is:

- (A) 490.7 nm
- (B) 194.8 nm
- (C) 913.3 nm
- (D) 121.8 nm

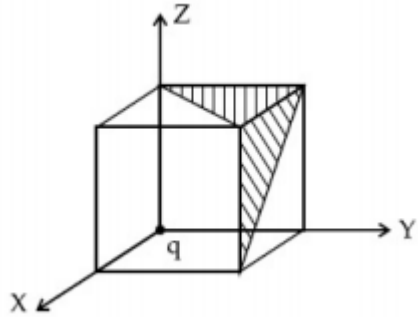
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Solution:

$$\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$
$$\lambda = 121.8 \text{ nm}$$

Q14: A charge 'q' is placed at one corner of a cube as shown in figure. The flux of electrostatic field $\vec{E}$ through the shaded area is:



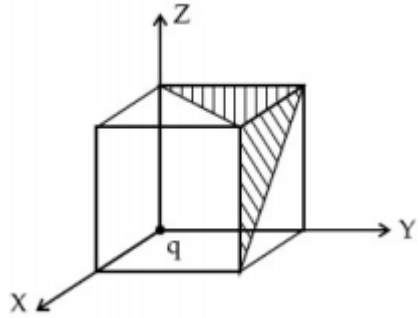
(A) $\frac{q}{24\epsilon_0}$

(B) $\frac{q}{4\epsilon_0}$

(C) $\frac{q}{48\epsilon_0}$

(D) $\frac{q}{8\epsilon_0}$

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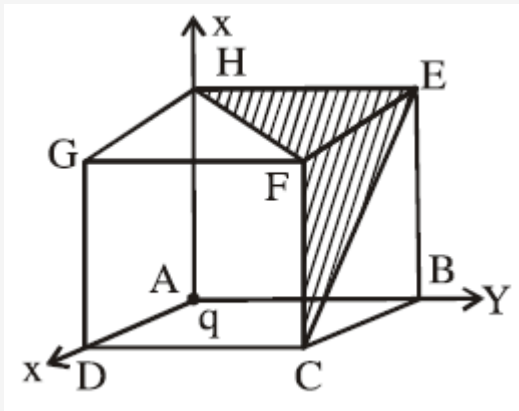
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Solution:



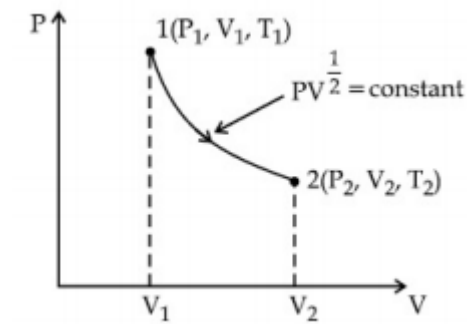
$$\text{Flux through cube} = \frac{q}{8\epsilon_0}$$

Flux through surfaces ABEH, ADGH, ABCD will be zero

$$\begin{aligned} \phi(EFGH) &= \phi(DCFG) = \phi(EBCF) = \frac{1}{3} \left(\frac{q}{8\epsilon_0} \right) \\ &= \frac{q}{24\epsilon_0} \end{aligned}$$

Q15: Thermodynamic process is shown below on a P-V diagram for one mole of an ideal gas.

If $V_2 = 2V_1$ then the ratio of temperature $\frac{T_2}{T_1}$ is:



(A) $\frac{1}{\sqrt{2}}$

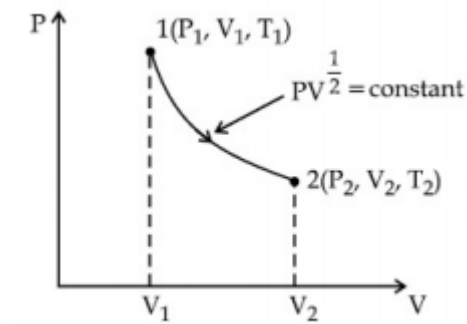
(B) $\sqrt{2}$

(C) $\frac{1}{2}$

(D) 2

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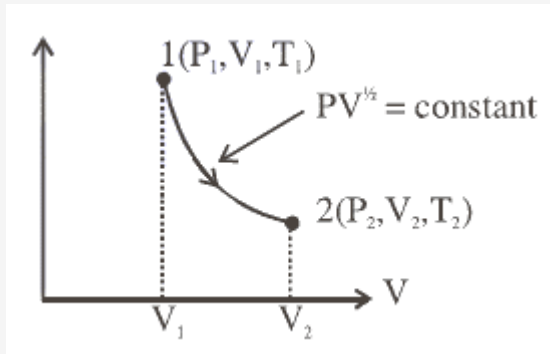
(A) $\frac{1}{\sqrt{2}}$

(B) $\sqrt{2}$

(C) $\frac{1}{2}$

(D) 2

Solution:



$$PV^{\frac{1}{2}} = c$$

$$\frac{nRT}{V} V^{\frac{1}{2}} = c$$

$$T = c^1 V^{\frac{1}{2}}$$

$$\frac{T_2}{T_1} = \left(\frac{V_2}{V_1} \right)^{\frac{1}{2}} = \left(\frac{2V_1}{V_1} \right)^{\frac{1}{2}}$$

$$\frac{T_2}{T_1} = \sqrt{2}$$

Q16: The stopping potential for electrons emitted from a photosensitive surface illuminated by light of wavelength 491 nm is 0.710 V. When the incident wavelength is changed to a new value, the stopping potential is 1.43 V. The new wavelength is:

- (A) 329 nm
- (B) 400 nm
- (C) 382 nm
- (D) 309 nm

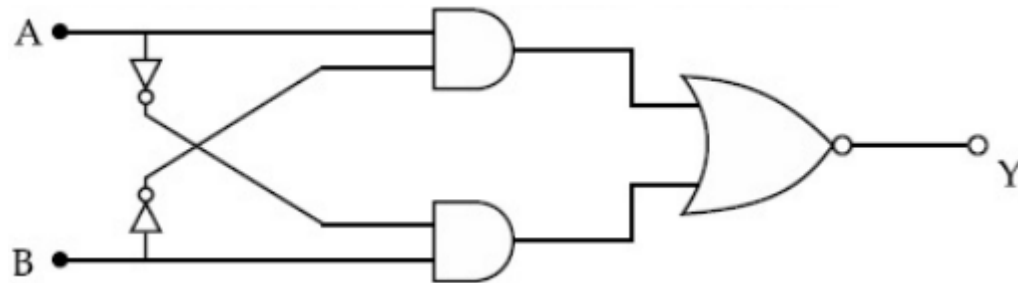
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Solution:

$$\begin{aligned}\frac{hc}{\lambda} &= \phi + eV_s \\ \frac{1240}{491} &= \phi + 0.71 \dots\dots (1) \\ \frac{1240}{\lambda} &= \phi + 1.43 \dots\dots (2) \\ \therefore \lambda &= 382 \text{ nm}\end{aligned}$$

Q17: The truth table for the following logic circuit is:



(A)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(C)

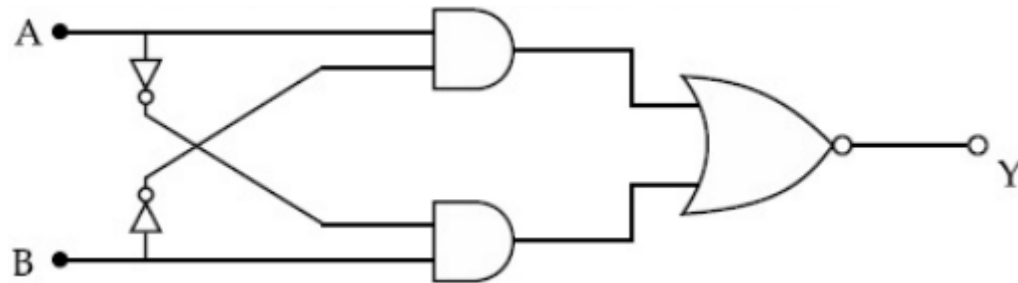
A	B	Y
0	0	0

0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

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(A)

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0	0	1
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(B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(C)

A	B	Y
0	0	0

0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

Solution:

$$y = \overline{(A\bar{B} + \bar{A}B)}$$

$$y = \overline{(A\bar{B}) \cdot \overline{\bar{A}B}}$$

$$y = (\bar{A} + B) \cdot (A + \bar{B})$$

$$y = \bar{A} \cdot A + \bar{A} \cdot \bar{B} + A \cdot B + B\bar{B}$$

$$y = AB + \bar{A}\bar{B}$$

A	B	$Y = AB + \bar{A}\bar{B}$
0	0	1
0	1	0
1	0	0
1	1	1

Q18: An LCR circuit contains resistance of $110\ \Omega$ and a supply of 220 V at 300 rad/ s angular frequency. If only capacitance is removed from the circuit, current lags behind the voltage by 45° . If on the other hand, only inductor is removed the current leads by 45° with the applied voltage. The rms current flowing in the circuit will be:

- (A) 2.5 A
- (B) 1 A
- (C) 1.5 A
- (D) 2 A

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Solution:

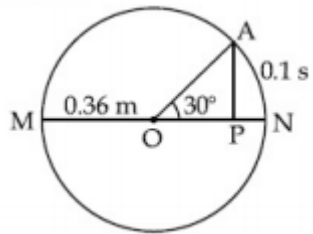
$$\tan 45^\circ = \frac{1}{\omega CR} = \frac{\omega L}{R} \Rightarrow X_L = X_C$$

$\Rightarrow$ Resonance

$$i = \frac{V}{R}$$

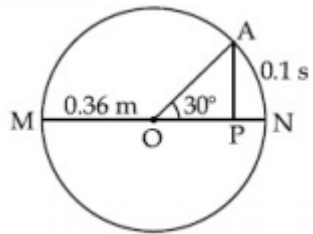
$$= \frac{220}{110} = 2A$$

Q19: The point A moves with a uniform speed along the circumference of a circle of radius 0.36 m and covers 30° in 0.1 s. The perpendicular projection 'P' from 'A' on the diameter MN represents the simple harmonic motion of 'P'. The restoration force per unit mass when P touches M will be:



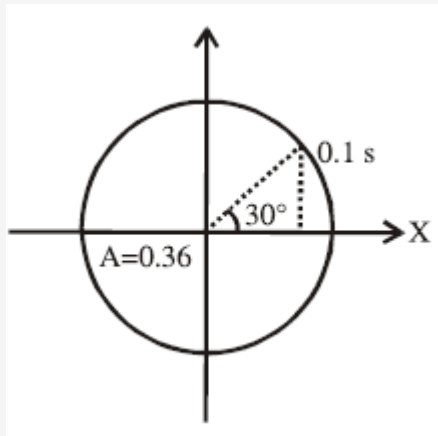
- (A) 0.49 N/kg
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- (B) 50 N/kg
- (C) 9.87 N/kg
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Solution:



$$30^\circ \rightarrow 0.1\text{ s}$$

$$360^\circ \rightarrow 1.2\text{ s} = T$$

$$\omega = \frac{2\pi}{T} = \frac{5\pi}{3}$$

$$\text{At M, } F = m\omega^2 A$$

$$\Rightarrow \frac{F}{m} = \omega^2 A = \left(\frac{5\pi}{3}\right)^2 \times 0.36 = 9.87\text{ N/kg}$$

Q20: If e is the electronic charge, c is the speed of light in free space and h is Planck's constant, the quantity $\frac{1}{4\pi\epsilon_0} \frac{|e|^2}{hc}$ has dimensions of:

- (A) $[MLT^0]$
- (B) $[LC^{-1}]$
- (C) $[MLT^{-1}]$
- (D) $[M^0L^0T^0]$

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- (C) $[MLT^{-1}]$
- (D) $[M^0 L^0 T^0]$

Solution:

$$F = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$E = \frac{hc}{\lambda}$$

$$\left[\frac{e^2}{4\pi\epsilon_0} \times \frac{1}{hc} \right] = \frac{Fr^2}{E\lambda} = [M^0 L^0 T^0]$$

Q21: Two identical conducting spheres with negligible volume have 2.1 nC and -0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is _____ $\times 10^{-9} N$.

[Given : $4\pi\epsilon_0 = \frac{1}{9 \times 10^9} \text{SI unit}$]

Q21: Two identical conducting spheres with negligible volume have 2.1 nC and -0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is _____ $\times 10^{-9} N$.

Given : $4\pi\epsilon_0 = \frac{1}{9 \times 10^9} \text{ SI unit}$

36.00

Solution:

$$q = \frac{(2.1 - 0.1)}{2} \text{ nC} = 1 \text{ nC}$$

$$F = \frac{9 \times 10^9 \times 10^{-18}}{(0.5)^2} = 36 \times 10^{-9}$$

Q22: The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4%. Will be _____ %.

Q22: The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4%. Will be _____ %.

2.00

Solution:

$$v = \sqrt{\frac{T}{\mu}}$$

$$\frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta T}{T} = \frac{1}{2} \times 4\% = 2\%$$

Q23: The peak electric field produced by the radiation coming from the 8 W bulb at a distance of 10 m is $\frac{x}{10} \sqrt{\frac{\mu_0 c}{\pi}} \frac{V}{m}$. The efficiency of the bulb is 10% and it is a point source. The value of x is _____.

Q23: The peak electric field produced by the radiation coming from the 8 W bulb at a distance of 10 m is $\frac{x}{10} \sqrt{\frac{\mu_0 c}{\pi}} \frac{V}{m}$. The efficiency of the bulb is 10% and it is a point source. The value of x is _____.

2.00

Solution:

$$I = \frac{1}{2} c \epsilon_0 E_0^2$$

$$\frac{8}{4\pi \times 10^2} \times \frac{1}{2} = \frac{1}{4} \times c \times \frac{1}{\mu_0 c^2} \times E_0^2$$

$$E_0 = \frac{2}{10} \times \sqrt{\frac{\mu_0 c}{\pi}} \Rightarrow x = 2$$

Q24: If $\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$, the angle between $\vec{P}$ and $\vec{Q}$ is θ ($0^\circ < \theta < 360^\circ$). The value of ' θ ' will be _____ $^\circ$.

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Solution:

$$\begin{aligned} PQ \sin \theta &= -PQ \sin \theta \\ \Rightarrow \theta &= 180^\circ \end{aligned}$$

Q25: The initial velocity v_i required to project a body vertically upward from the surface of the earth to reach a height of $10R$, where R is the radius of the earth, may be described in terms of escape velocity v_e such that $v_i = \sqrt{\frac{x}{y}} \times v_e$. The value of x will be_____.

Q25: The initial velocity v_i required to project a body vertically upward from the surface of the earth to reach a height of $10R$, where R is the radius of the earth, may be described in terms of escape velocity v_e such that $v_i = \sqrt{\frac{x}{y}} \times v_e$. The value of x will be_____.

10.00

Solution:

$$\frac{-GMm}{11R} = \frac{-GMm}{R} + \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{20GM}{11R}} = \sqrt{\frac{10}{11}} \times \sqrt{\frac{2GM}{R}}$$

$$v = \sqrt{\frac{10}{11}} \times v_e$$

So, $x = 10$

Q26: Two particles having masses 4 g and 16 g respectively are moving with equal kinetic energies. The ratio of the magnitudes of their linear momentum is $n : 2$. The value of n will be_____.

Q26: Two particles having masses 4 g and 16 g respectively are moving with equal kinetic energies. The ratio of the magnitudes of their linear momentum is $n : 2$. The value of n will be _____.

1.00

Solution:

$$\frac{p_1^2}{2 \times 4} = \frac{p_2^2}{2 \times 16}$$
$$\frac{p_1}{p_2} = \frac{1}{2}$$

Q27: A reversible heat engine converts one-fourth of the heat input into work. When the temperature of the sink is reduced by 52 K, its efficiency is doubled. The temperature in Kelvin of the source will be_____.

Q27: A reversible heat engine converts one-fourth of the heat input into work. When the temperature of the sink is reduced by 52 K, its efficiency is doubled. The temperature in Kelvin of the source will be_____.

208

Solution:

$$\eta = \frac{1}{4} = 1 - \frac{T_2}{T_1}$$

$$\frac{T_2}{T_1} = \frac{3}{4}$$

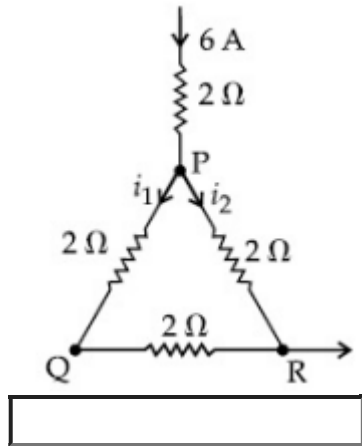
$$\frac{T_2 - 52}{T_1} = \frac{1}{2}$$

$$\frac{T_2}{T_1} - \frac{52}{T_1} = \frac{1}{2}$$

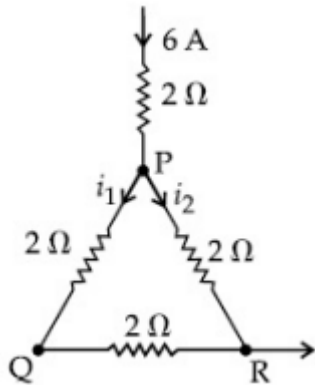
$$\frac{3}{4} - \frac{1}{2} = \frac{52}{T_1}$$

$$T_1 = 52 \times 4 = 208K$$

Q28: A current of 6 A enters one corner P of an equilateral triangle PQR having 3 wires of resistance $2\ \Omega$ each and leaves by the corner R. The currents i_1 in ampere is _____.



Q28: A current of 6 A enters one corner P of an equilateral triangle PQR having 3 wires of resistance $2\ \Omega$ each and leaves by the corner R. The currents i_1 in ampere is_____.



2.00

Solution:

By KCL at P

$$I_1 + I_2 = 6 \quad \dots(i)$$

By KVL For loop PQRP

$$-2I_1 - 2I_1 + 2I_2 = 0$$

$$\Rightarrow 2I_1 - I_2 = 0 \quad \dots(ii)$$

From equation (i) and (ii)

$$I_1 + I_2 + 2I_1 - I_2 = 6 + 0$$

$$\Rightarrow 3I_1 = 6$$

$$\Rightarrow I_1 = 2A$$

Q29: Two small spheres each of mass 10 mg are suspended from a point by threads 0.5 m long. They are equally charged and repel each other to a distance of 0.20 m. The charge on each of the sphere is $\frac{a}{21} \times 10^{-8} C$. The value of 'a' will be_____.

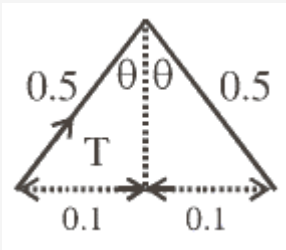
[Given $g = 10 \text{ ms}^{-2}$]

Q29: Two small spheres each of mass 10 mg are suspended from a point by threads 0.5 m long. They are equally charged and repel each other to a distance of 0.20 m. The charge on each of the sphere is $\frac{a}{21} \times 10^{-8} \text{ C}$. The value of 'a' will be_____.

[Given $g = 10 \text{ ms}^{-2}$]

20.00

Solution:



$$T \cos \theta = mg = 10 \times 10^{-6} \times 10 = 10^{-4}$$

$$T \sin \theta = \frac{9 \times 10^9 \times q^2}{0.04} = F$$

$$\tan \theta = \frac{0.1}{\sqrt{0.24}} = \frac{F}{mg}$$

$$q = \frac{2\sqrt{10}}{3\sqrt{\sqrt{24}}} \times 10^{-8}$$

$$0.95 \times 10^{-8} = \frac{a}{21} \times 10^{-8}$$

$$a = 20$$

Q30: The wavelength of an X-ray beam is $10 \text{ \AA}$. The mass of a fictitious particle having the same energy as that of the X-ray photons is $\frac{x}{3}h$ kg. The value of x is_____.

(h = Planck's constant)

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(h = Planck's constant)

10.00

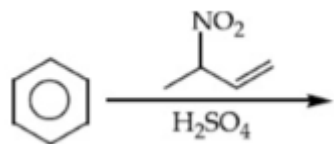
Solution:

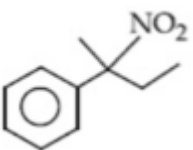
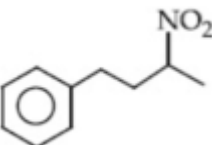
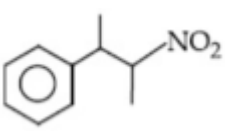
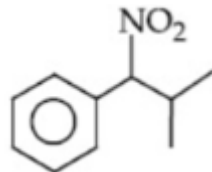
$$\frac{hc}{\lambda} = mc^2$$

$$m = \frac{h}{c\lambda}$$

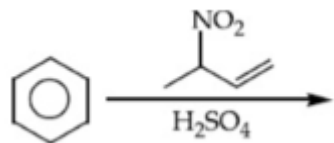
$$m = \frac{h}{3 \times 10^8 \times 10 \times 10^{-10}} = \frac{10}{3}h$$

Q31: The major product of the following reaction is:

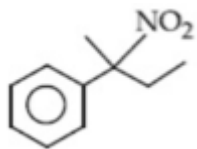


- (A) 
- (B) 
- (C) 
- (D) 

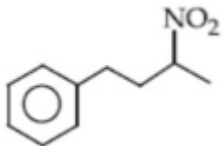
Q31: The major product of the following reaction is:



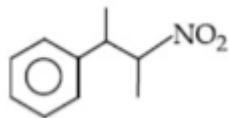
(A)



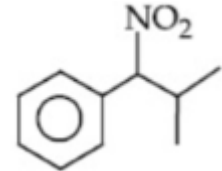
(B)



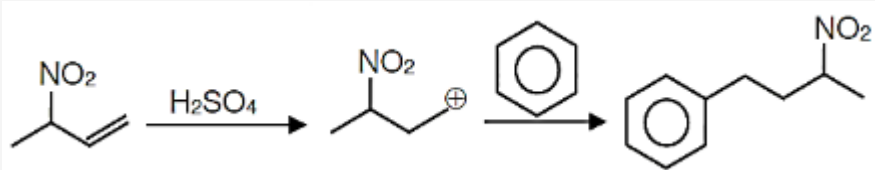
(C)



(D)



Solution:



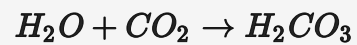
Q32: Water does not produce CO on reacting with

- (A) CH_4
- (B) C
- (C) C_3H_8
- (D) CO_2

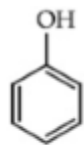
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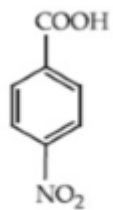
Solution:



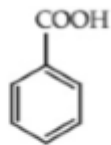
Q33: The correct order of acid character of the following compounds is:



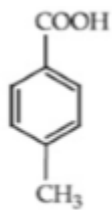
I



II



III



IV

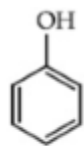
(A) IV > III > II > I

(B) II > III > IV > I

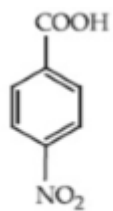
(C) III > II > I > IV

(D) I > II > III > IV

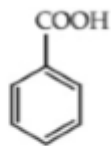
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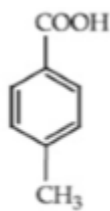
I



II



III



IV

(A) IV > III > II > I

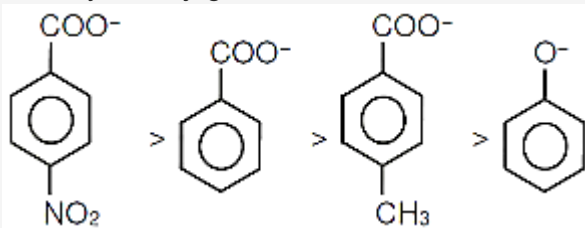
(B) II > III > IV > I

(C) III > II > I > IV

(D) I > II > III > IV

Solution:

Stability of conjugate bases :



Hence, acidic strength order will be

K_a : (ii) > (iii) > (iv) > (i)

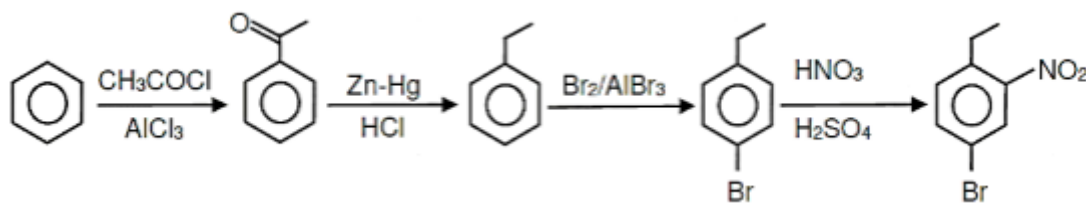
Q34: The correct sequence of reagents used in the preparation of 4-bromo-2-nitroethyl benzene from benzene is

- (A) $\text{CH}_3\text{COCl}/\text{AlCl}_3$, $\text{Zn-Hg}/\text{HCl}$, $\text{Br}_2/\text{AlBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$
- (B) $\text{HNO}_3/\text{H}_2\text{SO}_4$, $\text{Br}_2/\text{AlCl}_3$, $\text{CH}_3\text{COCl}/\text{AlCl}_3$, $\text{Zn-Hg}/\text{HCl}$
- (C) $\text{Br}_2/\text{AlBr}_3$, $\text{CH}_3\text{COCl}/\text{AlCl}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, Zn/HCl
- (D) $\text{CH}_3\text{COCl}/\text{AlCl}_3$, $\text{Br}_2/\text{AlBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, Zn/HCl

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- (B) $\text{HNO}_3/\text{H}_2\text{SO}_4$, $\text{Br}_2/\text{AlBr}_3$, $\text{CH}_3\text{COCl}/\text{AlCl}_3$, $\text{Zn-Hg}/\text{HCl}$
- (C) $\text{Br}_2/\text{AlBr}_3$, $\text{CH}_3\text{COCl}/\text{AlCl}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, Zn/HCl
- (D) $\text{CH}_3\text{COCl}/\text{AlCl}_3$, $\text{Br}_2/\text{AlBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, Zn/HCl

Solution:



Q35: Which of the following compound is added to the sodium extract before addition of silver nitrate for testing of halogens?

- (A) Ammonia
- (B) Hydrochloric acid
- (C) Nitric acid
- (D) Sodium hydroxide

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- (A) Ammonia
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- (D) Sodium hydroxide

Solution:

Sodium carbonate extract is acidified with HNO_3 only in identification of halide.

Q36: The major components of German Silver are

- (A) Ge, Cu and Ag
- (B) Cu, Zn and Ni
- (C) Zn, Ni and Ag
- (D) Cu, Zn and Ag

Q36: The major components of German Silver are

- (A) Ge, Cu and Ag
- ☒ (B) Cu, Zn and Ni
- (C) Zn, Ni and Ag
- (D) Cu, Zn and Ag

Solution:

Cu : 50% ; Zn : 20% ; Ni : 30% ; (Approximates)

Q37: The method used for the purification of Indium is

- (A) Van Arkel method
- (B) Liquation
- (C) Vapour phase refining
- (D) Zone refining

Q37: The method used for the purification of Indium is

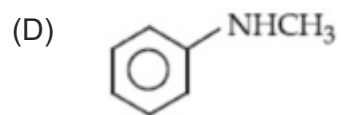
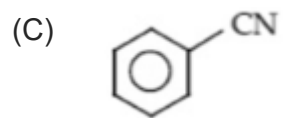
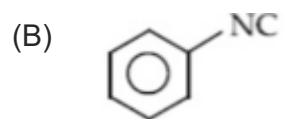
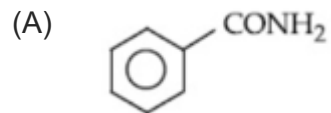
- (A) Van Arkel method
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- (D) Zone refining

Solution:

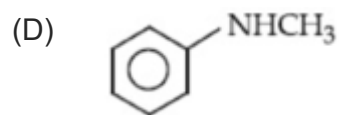
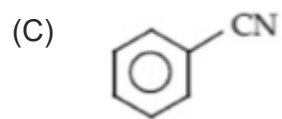
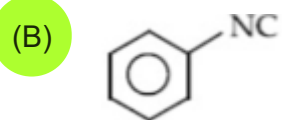
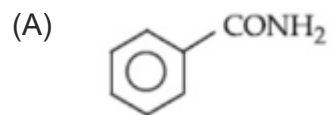
Zone refining

Principle: An Impure molten metal on gradual cooling will deposit crystal of pure metal while the impurities will be left in remaining molten part of molten metal. Impurities are more soluble in molten metal than solid metal. Used when metal of very high purity is needed. Example : Si, Ge, B, Ga, In.

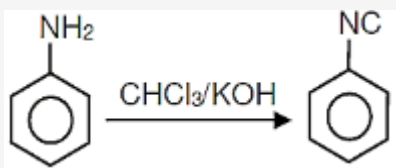
Q38: carbylamines test is used to detect the presence of primary amino group in an organic compound. Which of the following compound is formed when this test is performed with aniline?



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Solution:



Q39: Which one of the following statements is FALSE for hydrophilic sols?

- (A) Their viscosity is of the order of that of H_2O
- (B) They do not require electrolytes for stability
- (C) These sols are reversible in nature
- (D) The sols cannot be easily coagulated

Q39: Which one of the following statements is FALSE for hydrophilic sols?

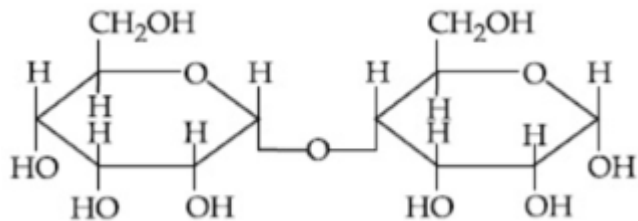
- (A) Their viscosity is of the order of that of H_2O
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Solution:

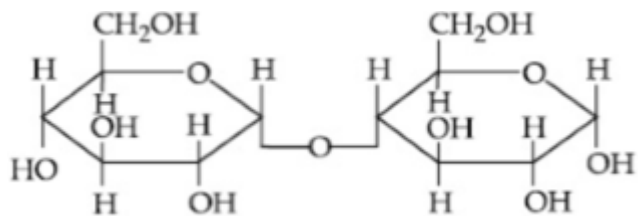
Hydrophilic colloids are water-loving colloids. The colloid particles are attracted toward the water. They are also called reversible sols. Their surface tension is usually lower than water, due to the breakdown of cohesive forces of the water molecules at the surface. The viscosity of the hydrophilic sols is much higher than that of the dispersion medium.

Q40: Which of the following is correct structure of α -anomer of maltose?

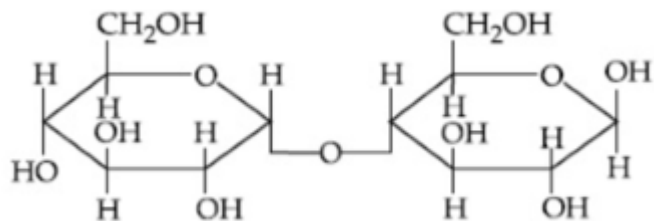
(A)



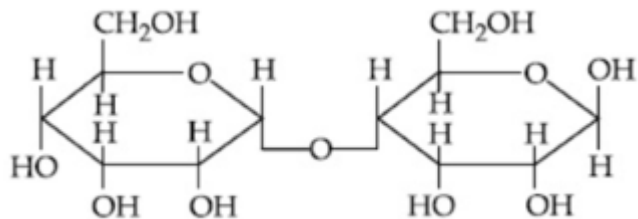
(B)



(C)

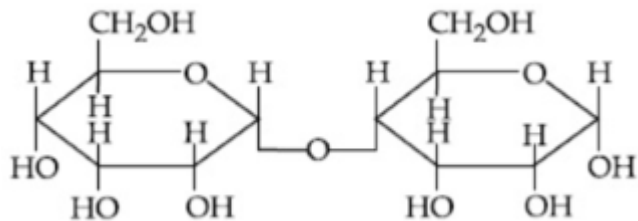


(D)

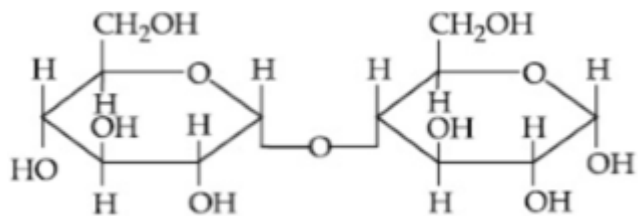


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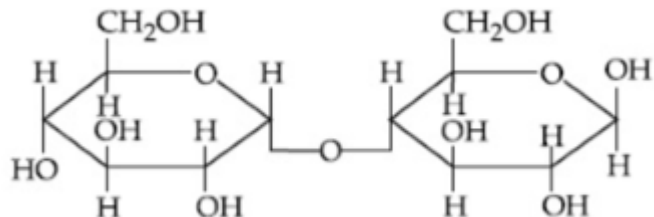
(A)



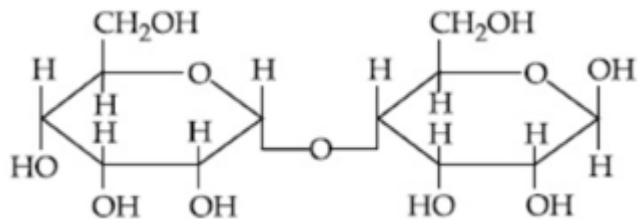
(B)



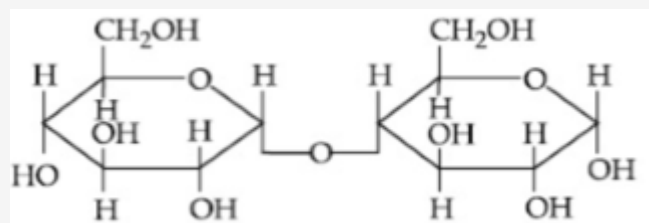
(C)



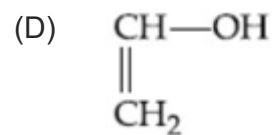
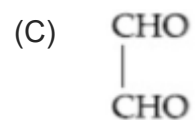
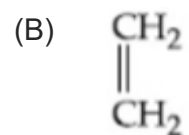
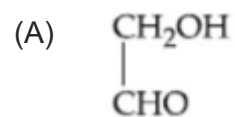
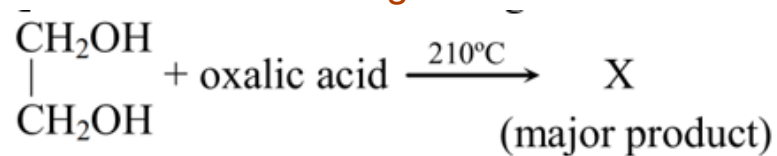
(D)



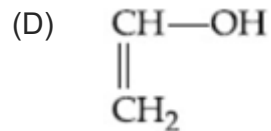
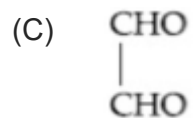
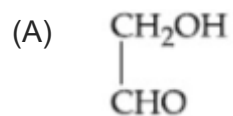
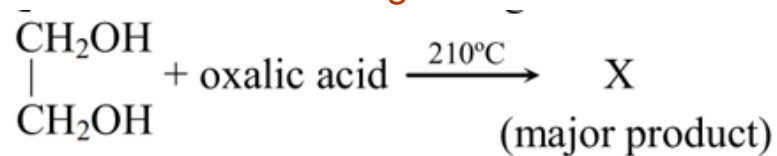
Solution:



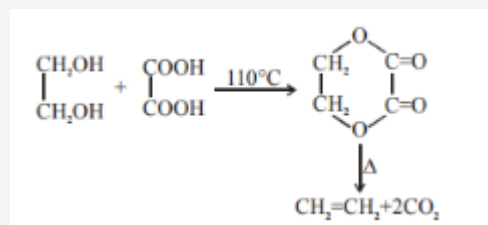
Q41: What is 'X' in the given reaction?



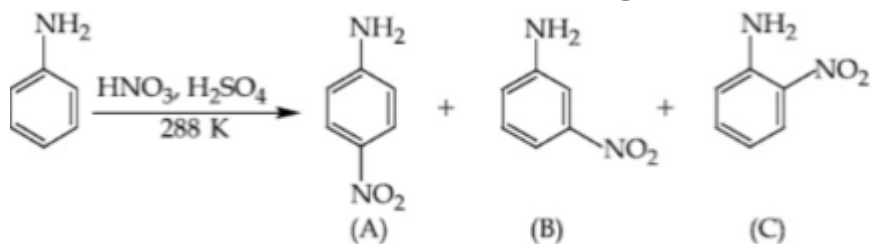
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Solution:

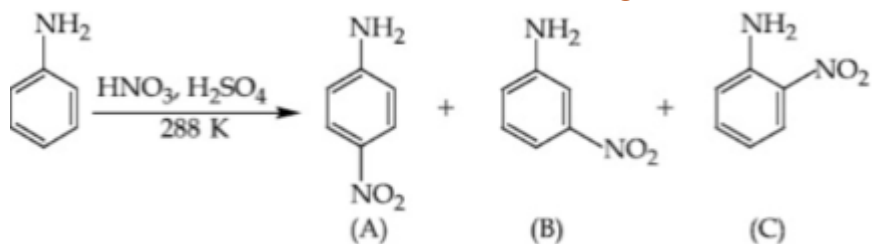


Q42: Correct statement about the given chemical reaction is:



- (A) Reaction is possible and compound (B) will be the major product
- (B) Reaction is possible and compound (A) will be major product
- (C) group is *ortho* and *para* directive, so product (B) is not possible
- (D) The reaction will form sulphonated product instead of nitration

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- (B) Reaction is possible and compound (A) will be major product
- (C) group is *ortho* and *para* directive, so product (B) is not possible
- (D) The reaction will form sulphonated product instead of nitration

Solution:

In nitration of aniline, Due to formation of anilinium ion, meta product is also formed.

Product formed are :

para $\rightarrow$ 51%

meta $\rightarrow$ 49%

ortho $\rightarrow$ 2%

Q43: The solubility of Ca(OH)_2 in water is:

[Given: The solubility product of Ca(OH)_2 in water = 5.5×10^{-6}]

- (A) 1.11×10^{-6}
- (B) 1.77×10^{-6}
- (C) 1.77×10^{-2}
- (D) 1.11×10^{-2}

Q43: The solubility of $\text{Ca}(\text{OH})_2$ in water is:

[Given: The solubility product of $\text{Ca}(\text{OH})_2$ in water = 5.5×10^{-6}]

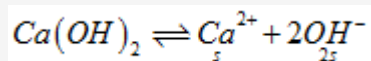
(A) 1.11×10^{-6}

(B) 1.77×10^{-6}

(C) 1.77×10^{-2}

(D) 1.11×10^{-2}

Solution:



$$K_{sp} = 4S^3 = 5.5 \times 10^{-6}$$

$$S = \left(\frac{5.5 \times 10^{-6}}{4} \right)^{1/3} = (1.375 \times 10^{-6})^{1/3}$$
$$= 1.110 \times 10^{-2} M$$

Q44: Given below are two statements:

Statement I: The pH of rain water is normally ~ 5.6 .

Statement II: If the pH of rain water drops below 5.6, it is called acid rain.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Statement I is true but statement II is false
- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are true

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- (C) Statement I is false but Statement II is true
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Solution:

pH range of rain is between 5 – 5.6 approximately and when pH of rain water becomes less than 5.6 ($\text{pH} < 5.6$) we call it as acid rain. So both statements are true.

Q45: Given below are two statements:

Statement I: α and β forms of sulphur can change reversibly between themselves with slow heating or slow cooling.

Statement II: At room temperature the stable crystalline form of sulphur is monoclinic sulphur.

In the light of the above statements, choose the correct answer from the options given below

- (A) Statement I is true but statement II is false
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Solution:

Statement – 1 : α – sulphur and β – sulphur are stable at 393K. So by applying some heat and cold both would be interconvertible. So statement one is true.

Statement – 2 : Most stable form of sulphur is rhombic, so statement two is false.

Correct option: Statement one is true but statement two is false

Q46: In which of the following order the given complex ions are arranged correctly with respect to their decreasing spin only magnetic moment?

(i) $[\text{FeF}_6]^{3-}$ (ii) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (iii) $[\text{NiCl}_3]^{2-}$ (iv) $[\text{Cu}(\text{NH}_3)_4]^{2+}$

(A) (i) > (iii) > (iv) > (ii)

(B) (ii) > (i) > (iii) > (iv)

(C) (iii) > (iv) > (ii) > (i)

(D) (ii) > (iii) > (i) > (iv)

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(B) (ii) > (i) > (iii) > (iv)

(C) (iii) > (iv) > (ii) > (i)

(D) (ii) > (iii) > (i) > (iv)

Solution:

Compound	EC	No. of unpaired electron
(i) $[\text{FeF}_6]^{3-}$	$\text{Fe}^{3+} \Rightarrow 3d^5$	5
(ii) $[\text{Co}(\text{CN})_6]^{3-}$	$\text{Co}^{3+} \Rightarrow 3d^6$	0
(iii) $[\text{NiCl}_4]^{2-}$	$\text{Ni}^{2+} \Rightarrow 3d^8$	2
(iv) $[\text{Cu}(\text{NH}_3)_4]^{2+}$	$\text{Cu}^{2+} \Rightarrow 3d^9$	1

Q47: Given below are two statements:

Statement I: The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH

Statement II: The dimethyl glyoxime is a bidentate neutral ligand.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both Statement I and statement II are false
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Q47: Given below are two statements:

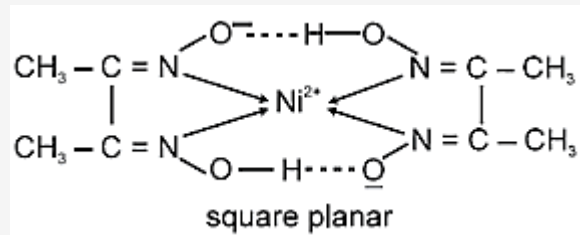
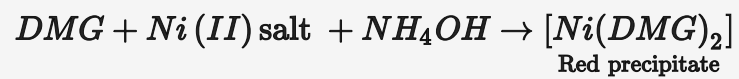
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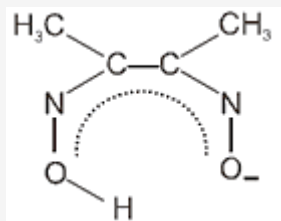
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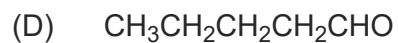
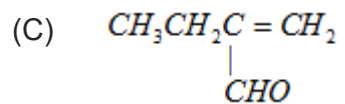
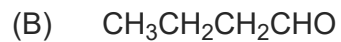
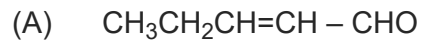
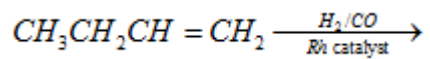
Solution:



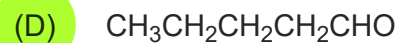
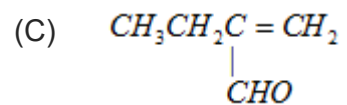
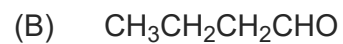
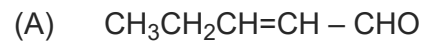
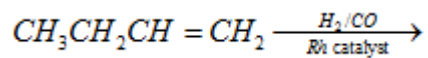
DMG structure:



Q48: The major product of the following reaction is:



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Solution:

This is oxo process to prepare aldehyde from alkene

Q49: The correct order of bond dissociation enthalpy of halogens is

- (A) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$
- (B) $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$
- (C) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- (D) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$

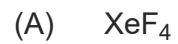
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- (C) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- (D) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$

Solution:

Enthalpy of dissociation (X_2) / kJ mol^{-1} ($\text{Cl}-\text{Cl} > \text{Br}-\text{Br} > \text{F}-\text{F} > \text{I}-\text{I}$)	158.8	242.6	192.8	151.1
σ Smaller enthalpy of dissociation of F_2 is due to relatively larger electron – electron repulsion among the lone pairs in F_2 molecule.				

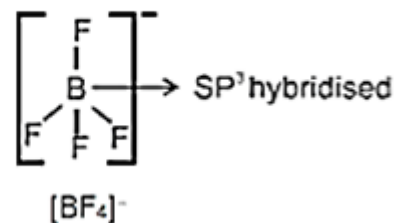
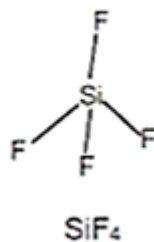
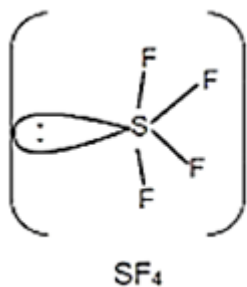
Q50: Which among the following species has unequal bond lengths?



Q50: Which among the following species has unequal bond lengths?

- (A) XeF_4
- (B) SiF_4
- (C) SF_4
- (D) BF_4^-

Solution:



Q51:

Copper reduces NO_3^- into NO and NO_2 depending upon the concentration of HNO_3 in solution. (Assuming fixed $[\text{Cu}^{2+}]$ and $P_{\text{NO}} = P_{\text{NO}_2}$), the HNO_3 concentration at which the thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same is 10^x M. The value of $2x$ is (Rounded-off to the nearest integer)

[Given, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34$ V, $E^\circ_{\text{NO}_3^-/\text{NO}} = 0.96$ V, $E^\circ_{\text{NO}_3^-/\text{NO}_2} = 0.79$ V and at 298 K, $\frac{RT}{F} (2.303) = 0.059$]

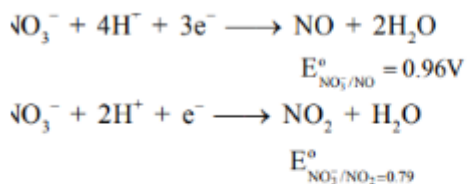
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Solution:

If the partial pressure of NO and NO₂ gas is taken as 1 bar, then Answer is 4, else the question is bonus.



Let $[\text{HNO}_3] = y \Rightarrow [\text{H}^+] = y$ and $[\text{NO}_3^-] = y$ for same thermodynamic tendency

$$E_{\text{NO}_3^-/\text{NO}} = E_{\text{NO}_3^-/\text{NO}_2}$$

$$\text{or, } E_{\text{NO}_3^-/\text{NO}}^\circ - \frac{0.059}{3} \log \frac{P_{\text{NO}}}{y \times y^4}$$

$$= E_{\text{NO}_3^-/\text{NO}_2}^\circ - \frac{0.059}{1} \log \frac{P_{\text{NO}_2}}{y \times y^2}$$

$$\text{or, } 0.96 - \frac{0.059}{3} \log \frac{P_{\text{NO}}}{y^5} = 0.79 - \frac{0.059}{1} \log \frac{P_{\text{NO}_2}}{y^3}$$

$$\text{or, } 0.17 = -\frac{0.059}{1} \log \frac{P_{\text{NO}_2}}{y^3} + \frac{0.059}{3} \log \frac{P_{\text{NO}}}{y^5}$$

$$0.17 = -\frac{0.0591}{1} \log \frac{P_{\text{NO}_2}}{y^3} + \frac{0.0591}{3} \log \frac{P_{\text{NO}}}{y^5}$$

$$0.17 = -\frac{0.0591}{3} \log \frac{P_{\text{NO}_2}^3}{y^9} + \frac{0.0591}{3} \log \frac{P_{\text{NO}}}{y^5}$$

$$0.17 = \frac{0.0591}{3} \left[\log \frac{P_{\text{NO}}}{y^5} - \log \frac{P_{\text{NO}_2}^3}{y^9} \right]$$

$$0.17 = \frac{0.0591}{3} \left[\log \frac{P_{\text{NO}}}{y^5} \times \frac{y^9}{P_{\text{NO}_2}^3} \right]$$

Assume $P_{\text{NO}} = P_{\text{NO}_2} = 1 \text{ bar}$

$$\frac{0.17 \times 3}{0.059} = \log y^4 = 8.644$$

$$\log y = \frac{8.644}{4}$$

$$\log y = 2.161$$

$$y = 10^{2.16}$$

$$\therefore 2x = 2 \times 2.161 = 4.322$$

Q52: If a compound AB dissociates to the extent of 75% in an aqueous solution, the molality of the solution which shows a 25K rise in the boiling point of the solution is _____ molal.

[$K_b = 0.52\text{K kg mol}^{-1}$]

Q52: If a compound AB dissociates to the extent of 75% in an aqueous solution, the molality of the solution which shows a 25K rise in the boiling point of the solution is _____ molal.

$[K_b = 0.52\text{K kg mol}^{-1}]$

3.00

Solution:

$$\Delta T_b = iK_b m$$

$$i = 1 + (n - 1)\alpha$$

$$= 1 + [2 - 1]0.75$$

$$= 1.75$$

$$\Delta T_b = iK_b m$$

$$\Delta T_b = 2.5\text{K}$$

$$2.5 = 1.75[0.52]m$$

$$m = 2.74$$

So Ans. is 3

Q53: Five moles of an ideal gas at 293K is expanded isothermally from an initial pressure of 2.1 MPa to 1.3 Mpa against at constant external pressure 4.3 MPa. The heat transferred in this process is _____ kJ mol⁻¹. (Rounded-off to the nearest integer)
[Use $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

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[Use R = 8.314 J mol⁻¹K⁻¹]

15.00

Solution:

$$\begin{aligned}
 V_f &= \frac{5 \times 8.314 \times 293}{1.3 \times 10^6} = 9369.23 \times 10^{-6} \\
 V_i &= \frac{5 \times 8.314 \times 293}{2.1 \times 10^6} = 5800 \times 10^{-6} \\
 W &= -P_{ext} (V_f - V_i) \\
 &= -4.3 \times 10^6 (9369.23 - 5800) \times 10^{-6} \text{ J mol}^{-1} \\
 &= -4.3 \times 3569.23 \text{ J mol}^{-1} \\
 &= -1547.7 \text{ J mol}^{-1} \\
 &= -15.34 \text{ kJ mol}^{-1} \\
 &= Q = -W \\
 Q &= 15 \text{ kJ mol}^{-1}
 \end{aligned}$$

Q54: The unit cell of copper corresponds to a face centered cube of edge length 3.5×10^{-10} m with one copper atom at each lattice point. The calculated density of copper in kg/m^3 is ____.

[Molar mass of Cu : 63.54 g; Avogadro Number = 6.022×10^{23}]

Q54: The unit cell of copper corresponds to a face centered cube of edge length $3.56 \text{ \AA}$ with one copper atom at each lattice point. The calculated density of copper in kg/m^3 is _____.
[Molar mass of Cu : 63.54 g ; Avogadro Number = 6.022×10^{23}]

9076.00

Solution:

FCC,

$$\begin{aligned} d &= \frac{Z \times M}{N_A \times a^3} \\ &= \frac{4 \times 63.54}{6.022 \times 10^{23} \times (3.56 \times 10^{-10})^3} \\ &= 9076 \text{ kg/m}^3 \end{aligned}$$

Q55: Among the following number of metal/s which can be used as electrodes in the photoelectric cell is _____(Integer answer)

(A) Li (B) Na (C) Rb (D) Cs

Q55: Among the following number of metal/s which can be used as electrodes in the photoelectric cell is _____(Integer answer)

(A) Li (B) Na (C) Rb (D) Cs

1.00

Solution:

Cs is used as electrode in photoelectric cell.

Q56:

Calculate wavelength of a photon in terms of 10^{-7} m, which has energy equal to $5 \times 10^{-19} \text{ J}$.
Report answer after rounding off to two digits of decimal. ($h = 6.626 \times 10^{-34} \text{ Js}$)

Q56:

Calculate wavelength of a photon in terms of 10^{-7} m, which has energy equal to 5×10^{-19} J. Report answer after rounding off to two digits of decimal. ($h = 6.626 \times 10^{-34}$ Js)

3.98

Solution:

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$$

$$\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{5 \times 10^{-19}}$$

$$= 3.9756 \times 10^{-7} m$$

Q57: Consider titration of NaOH solution versus 1.25M oxalic acid solution. At the end point following burette readings were obtained.

(i) 4.5 mL (ii) 4.5 mL (iii) 4.4 mL

(iv) 4.4 mL (v) 4.4 mL

If the volume of oxalic acid taken was 10.0 mL then the molarity of the NaOH solution is _____ M. (Rounded-off the nearest integer)

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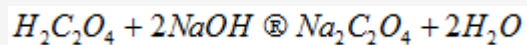
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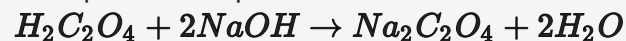
6.00

Solution:



$$VF = 2 \quad VF = 1$$

Mili eq. of = Mili eq. of NaOH



$$2 \times 1.25 \times 10 = 1 \times M \times 4.4$$

$$M = 5.68$$

Ans. 6

Q58: The number of compound/s given below which contain/s – COOH group is____. (Integer answer)

(A) Sulphanilic acid (B) Picric acid

(C) Aspirin (D) Ascorbic acid

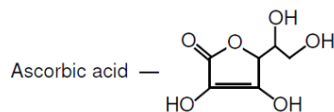
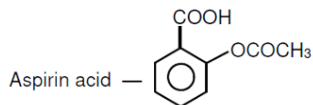
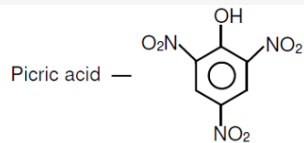
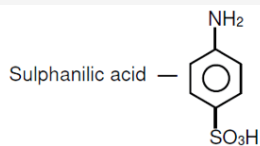
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(A) Sulphanilic acid (B) Picric acid

(C) Aspirin (D) Ascorbic acid

1.00

Solution:

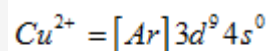
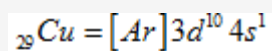


Q59: The spin only magnetic moment of a divalent ion in an aqueous solution (atomic number 29) is _____ BM. (round off your answer to the nearest integer)

Q59: The spin only magnetic moment of a divalent ion in an aqueous solution (atomic number 29) is _____ BM. (round off your answer to the nearest integer)

2.00

Solution:



so $n = 1$

$$\mu = \sqrt{n(n+2)}$$

$$= \sqrt{3} \text{ BM}$$

$$= 1.73 \text{ BM}$$

$$= 2 \text{ B.M}$$

Q60: The rate constant of a reaction increases by five times on increase in temperature from 27°C to 52°C. The value of activation energy in kJ mol⁻¹ is _____. (Rounded-off to the nearest integer)

[R = 8.314 J K⁻¹ mol⁻¹]

Q60: The rate constant of a reaction increases by five times on increase in temperature from 27°C to 52°C. The value of activation energy in kJ mol⁻¹ is _____. (Rounded-off to the nearest integer)

[R = 8.314 J K⁻¹ mol⁻¹]

52.00

Solution:

$$\log \left(\frac{K_2}{K_1} \right) = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log 5 = \frac{E_a}{2.303 \times 8.314} \left[\frac{1}{300} - \frac{1}{325} \right]$$

$$0.7 = \frac{E_a}{2.303 \times 8.314} \left[\frac{25}{300 \times 325} \right]$$

$$E_a = 0.7 \times 8.314 \times 2.303 \times 300 \times 13$$

$$E_a = 52271.697 \text{ J}$$

$$E_a = 52.27 \times 10^3 \text{ J} = 52 \text{ kJ}$$

Q61: The contrapositive of the statement “ If you will work, you will earn money” is:

- (A) To earn money, you need to work
- (B) If you will not earn money, you will not work
- (C) You will earn money, if you will not work
- (D) If you will earn money, you will work

Q61: The contrapositive of the statement “ If you will work, you will earn money” is:

- (A) To earn money, you need to work
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- (D) If you will earn money, you will work

Solution:

Contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$

$\Rightarrow$ If you will not earn money, you will not work.

Q62: If the curve $x^2 + 2y^2 = 2$ intersects the line $x + y = 1$ at two points P and Q , then the angle subtended by the line segment PQ at the origin is :

(A) $\frac{\pi}{2} + \tan^{-1} \left(\frac{1}{3} \right)$

(B) $\frac{\pi}{2} + \tan^{-1} \left(\frac{1}{4} \right)$

(C) $\frac{\pi}{2} - \tan^{-1} \left(\frac{1}{4} \right)$

(D) $\frac{\pi}{2} - \tan^{-1} \left(\frac{1}{3} \right)$

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(C) $\frac{\pi}{2} - \tan^{-1} \left(\frac{1}{4} \right)$

(D) $\frac{\pi}{2} - \tan^{-1} \left(\frac{1}{3} \right)$

Solution:

On homogenizing,

$$x^2 + 2y^2 - 2(x + y)^2 = 0$$

$$\Rightarrow -x^2 - 4xy = 0$$

$$\Rightarrow x^2 + 4xy = 0$$

Lines are $x = 0$ and $y = -\frac{x}{4}$

$$\therefore \text{Angle between lines} = \frac{\pi}{2} + \tan^{-1} \frac{1}{4}$$

Q63: Let A be a set of all 4-digit natural numbers whose exactly one digit is 7. Then the probability that a randomly chosen element of A leaves remainder 2 when divided by 5 is :

(A) $\frac{97}{297}$

(B) $\frac{122}{297}$

(C) $\frac{2}{9}$

(D) $\frac{1}{5}$

Q63: Let A be a set of all 4-digit natural numbers whose exactly one digit is 7. Then the probability that a randomly chosen element of A leaves remainder 2 when divided by 5 is :

(A) $\frac{97}{297}$

(B) $\frac{122}{297}$

(C) $\frac{2}{9}$

(D) $\frac{1}{5}$

Solution:

$$n(s) = n(\text{when 7 appears on thousands place})$$

$$+ n(7 \text{ does not appear on thousands place})$$

$$= 9 \times 9 \times 9 + 8 \times 9 \times 9 \times 3$$

$$= 33 \times 9 \times 9$$

$$n(E) = n(\text{last digit 7 and 7 appears once})$$

$$+ n(\text{last digit 2 when 7 appears once})$$

$$= 8 \times 9 \times 9 + (9 \times 9 + 8 \times 9 \times 2)$$

$$\therefore P(E) = \frac{8 \times 9 \times 9 + 9 \times 25}{33 \times 9 \times 9} = \frac{97}{297}$$

Q64: Let x denote the total number of one – one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one one functions from the set A to the set $A \times B$. Then:

- (A) $y = 91x$
- (B) $2y = 91x$
- (C) $y = 273x$
- (D) $2y = 273x$

Q64: Let x denote the total number of one – one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one one functions from the set A to the set $A \times B$. Then:

(A) $y = 91x$

(B) $2y = 91x$

(C) $y = 273x$

(D) $2y = 273x$

Solution:

$$x = {}^5C_3 \times 3! = 60$$

$$y = {}^{15}C_3 \times 3! = 15 \times 14 \times 13 = 30 \times 91$$

$$\therefore 2y = 91x$$

Q65: A hyperbola passes through the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and conjugate axes coincide with major and minor axes of the ellipse, respectively. If the product of their eccentricities is one, then the equation of the hyperbola is:

(A) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(B) $x^2 - y^2 = 9$

(C) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

(D) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

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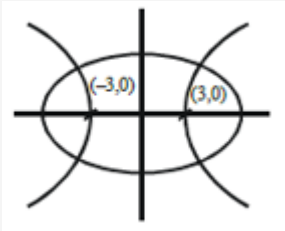
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(B) $x^2 - y^2 = 9$

(C) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

(D) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

Solution:



For ellipse $e_1 = \sqrt{1 - \frac{b^2}{a^2}} = \frac{3}{5}$

For hyperbola $e_2 = \frac{5}{3}$

Let hyperbola be

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$\therefore$ it passes through $(3, 0) \Rightarrow \frac{9}{a^2} = 1$

$$\Rightarrow a^2 = 9$$

$$\Rightarrow b^2 = a^2 (e^2 - 1)$$

$$= 9 \left(\frac{25}{9} - 1 \right) = 16$$

$\therefore$ hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$

Q66: The shortest distance between the line $x - y = 1$ and the curve $x^2 = 2y$ is:

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) 0

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(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) 0

Solution:

Let $P(x_0, y_0)$ be any point on the curve $x^2 = 2y$

Shortest distance between curves is always along common normal.

$$\left. \frac{dy}{dx} \right|_P = \text{slope of line} = 1$$

$$\Rightarrow x_0 = 1 \quad \therefore y_0 = \frac{1}{2}$$

$$\Rightarrow P\left(1, \frac{1}{2}\right)$$

$$\therefore \text{Shortest distance} = \left| \frac{1 - \frac{1}{2} - 1}{\sqrt{1^2 + 1^2}} \right| = \frac{1}{2\sqrt{2}}$$

Q67: If $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \, dx$, then

(A) $I_2 + I_4, (I_3 + I_5)^2, I_4 + I_6$ are in G.P.

(B) $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}$ are in A.P.

(C) $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}$ are in G.P.

(D) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in A.P.

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(C) $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}$ are in G.P.

(D) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in A.P.

Solution:

$$I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{n-2} x (\operatorname{cosec}^2 x - 1) \, dx$$

$$= -\frac{\cot^{n-1} x}{n-1} \Bigg]_{\frac{\pi}{4}}^{\frac{\pi}{2}} - I_{n-2}$$

$$= \frac{1}{n-1} - I_{n-2}$$

$$\Rightarrow I_n + I_{n-2} = \frac{1}{n-1}$$

$$\Rightarrow I_2 + I_4 = \frac{1}{3}$$

$$I_3 + I_5 = \frac{1}{4}$$

$$I_4 + I_6 = \frac{1}{5}$$

$\therefore \frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in A.P.

Q68: Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is:

(A) 2

(B) 4

(C) 1

(D) 3

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(A) 2

(B) 4

(C) 1

(D) 3

Solution:

$$\begin{array}{r}
 \alpha^2 - 6\alpha - 2 = 0 \\
 \alpha^{10} - 6\alpha^9 - 2\alpha^8 = 0 \\
 \beta^{10} - 6\beta^9 - 2\beta^8 = 0 \\
 \hline
 (\alpha^{10} - \beta^{10}) - 6(\alpha^9 - \beta^9) - 2(\alpha^8 - \beta^8) = 0 \\
 \Rightarrow a_{10} - 6a_9 - 2a_8 = 0 \\
 \Rightarrow \frac{a_{10} - 2a_8}{3a_9} = 2
 \end{array}$$

Q69: A function $f(x)$ is given by $f(x) = \frac{5^x}{5^x + 5}$, then the sum of the series $f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \dots + f\left(\frac{39}{20}\right)$

(A) $\frac{49}{2}$

(B) $\frac{29}{2}$

(C) $\frac{39}{2}$

(D) $\frac{19}{2}$

Q69: A function $f(x)$ is given by $f(x) = \frac{5^x}{5^x+5}$, then the sum of the series $f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \dots + f\left(\frac{39}{20}\right)$

(A) $\frac{49}{2}$

(B) $\frac{29}{2}$

(C) $\frac{39}{2}$

(D) $\frac{19}{2}$

Solution:

$$f(x) = \frac{5^x}{5^x+5} \quad f(2-x) = \frac{5}{5^x+5}$$

$$f(x) + f(2-x) = 1$$

$$\Rightarrow f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + \dots + f\left(\frac{39}{20}\right)$$

$$= \left(f\left(\frac{1}{20}\right) + f\left(\frac{39}{20}\right)\right) + \dots + \left(f\left(\frac{19}{20}\right) + f\left(\frac{21}{20}\right) + f\left(\frac{20}{20}\right)\right)$$

$$= 19 + \frac{1}{2} = \frac{39}{2}$$

Q70: The integral $\int \frac{e^{3\log_e 2x} + 5e^{2\log_e 2x}}{e^{4\log_e x} + 5e^{3\log_e x} - 7e^{2\log_e x}} dx, x > 0$, is equal to:

(where c is a constant of integration)

(A) $4\log_e |x^2 + 5x - 7| + c$

(B) $\log_e \sqrt{x^2 + 5x - 7} + c$

(C) $\frac{1}{4}\log_e |x^2 + 5x - 7| + c$

(D) $\log_e |x^2 + 5x - 7| + c$

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(C) $\frac{1}{4}\log_e |x^2 + 5x - 7| + c$

(D) $\log_e |x^2 + 5x - 7| + c$

Solution:

$$\begin{aligned} & \int \frac{e^{3\log_e 2x} + 5e^{2\log_e 2x}}{e^{4\log_e x} + 5e^{3\log_e x} - 7e^{2\log_e x}} dx, x > 0 \\ &= \int \frac{(2x)^3 + 5(2x)^2}{x^4 + 5x^3 - 7x^2} dx = \int \frac{4x^2(2x + 5)}{x^2(x^2 + 5x - 7)} dx \\ &= 4 \int \frac{d(x^2 + 5x - 7)}{(x^2 + 5x - 7)} = 4\log_e |x^2 + 5x - 7| + c \end{aligned}$$

Q71: In a group of 400 people, 160 are smokers and non-vegetarian; 100 are smokers and vegetarian and the remaining 140 are non-smokers and vegetarian. Their chances of getting a particular chest disorder are **35%**, **20%** and **10%** respectively. A person is chosen from the group at random and is found to be suffering from the chest disorder. The probability that the selected person is a smoker and non-vegetarian is:

(A) $\frac{14}{45}$

(B) $\frac{7}{45}$

(C) $\frac{8}{45}$

(D) $\frac{28}{45}$

Q71: In a group of 400 people, 160 are smokers and non-vegetarian; 100 are smokers and vegetarian and the remaining 140 are non-smokers and vegetarian. Their chances of getting a particular chest disorder are **35%**, **20%** and **10%** respectively. A person is chosen from the group at random and is found to be suffering from the chest disorder. The probability that the selected person is a smoker and non-vegetarian is:

(A) $\frac{14}{45}$

(B) $\frac{7}{45}$

(C) $\frac{8}{45}$

(D) $\frac{28}{45}$

Solution:

Consider the following events

A : Person chosen is a smoker and non - vegetarian.

B : Person chosen is a smoker and vegetarian.

C : Person chosen is a non – smoker and vegetarian.

E : Person chosen has a chest disorder

Given,

$$P(A) = \frac{160}{400}, P(B) = \frac{100}{400}, P(C) = \frac{140}{400}$$

$$P\left(\frac{E}{A}\right) = \frac{35}{100}, P\left(\frac{E}{B}\right) = \frac{20}{100}, P\left(\frac{E}{C}\right) = \frac{10}{100}$$

To find

$$\begin{aligned} P\left(\frac{A}{E}\right) &= \frac{P(A)P\left(\frac{E}{A}\right)}{P(A).P\left(\frac{E}{A}\right) + P(B).P\left(\frac{E}{B}\right) + P(C).P\left(\frac{E}{C}\right)} \\ &= \frac{\frac{160}{400} \times \frac{35}{100}}{\frac{160}{400} \times \frac{35}{100} + \frac{100}{400} \times \frac{20}{100} + \frac{140}{400} \times \frac{10}{100}} \\ &= \frac{28}{45} \end{aligned}$$

Q72: $\operatorname{cosec}\left[2\cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to:

(A) $\frac{65}{56}$

(B) $\frac{65}{33}$

(C) $\frac{75}{56}$

(D) $\frac{56}{33}$

Q72: $\operatorname{cosec}\left[2\cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to:

(A) $\frac{65}{56}$

(B) $\frac{65}{33}$

(C) $\frac{75}{56}$

(D) $\frac{56}{33}$

Solution:

$$\begin{aligned} & \operatorname{cosec}\left[2\tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{3}{4}\right)\right] \\ & \operatorname{cosec}\left[\tan^{-1}\left(\frac{5}{12}\right) + \tan^{-1}\left(\frac{3}{4}\right)\right] \\ & = \operatorname{cosec}\left[\tan^{-1}\left(\frac{56}{33}\right)\right] = \frac{65}{56} \end{aligned}$$

Q73: If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is:

(A) 3

(B) 1

(C) 4

(D) 2

Q73: If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is:

(A) 3

(B) 1

(C) 4

(D) 2

Solution:

$$\begin{aligned} A &= \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix} & AA^T &= I_2 \\ \Rightarrow \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix} \begin{bmatrix} 1 & \alpha \\ -\alpha & \beta \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 1 + \alpha^2 & \alpha - \alpha\beta \\ \alpha - \alpha\beta & \alpha^2 + \beta^2 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &\Rightarrow \alpha^2 = 0 \text{ and } \beta^2 = 1 \\ \therefore \alpha^4 + \beta^4 &= 1 \end{aligned}$$

Q74: The following system of linear equations

$$2x + 3y + 2z = 9$$

$$3x + 2y + 2z = 9$$

$$x - y + 4z = 8$$

- (A) does not have any solution
- (B) has infinitely many solutions
- (C) has a unique solution
- (D) has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^3 = 12$

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$$x - y + 4z = 8$$

- (A) does not have any solution
- (B) has infinitely many solutions
- (C) has a unique solution
- (D) has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^3 = 12$

Solution:

$$2x + 3y + 2z = 9 \quad \dots (i)$$

$$3x + 2y + 2z = 9 \quad \dots (ii)$$

$$x - y + 4z = 8 \quad \dots (iii)$$

$$(i) - (ii) \Rightarrow -x + y = 0 \Rightarrow x - y = 0$$

$$\text{From (iii)} \quad 4z = 8 \Rightarrow z = 2$$

$$\text{From (i)} \quad 2x + 3y = 5$$

$$\Rightarrow x = y = 1$$

$\therefore$ system has unique solution

Q75: The minimum of $f(x) = a^{a^x} + a^{1-a^x}$, where $a, x \in \mathbb{R}$ and $a > 0$, is equal to:

(A) $a + 1$

(B) $2\sqrt{a}$

(C) $2a$

(D) $a + \frac{1}{a}$

Q75: The minimum of $f(x) = a^{a^x} + a^{1-a^x}$, where $a, x \in \mathbb{R}$ and $a > 0$, is equal to:

(A) $a + 1$

(B) $2\sqrt{a}$

(C) $2a$

(D) $a + \frac{1}{a}$

Solution:

$$\begin{aligned} A.M. &\geq G.M. \\ &= a^{a^x} + \frac{a}{a^{a^x}} \geq 2\sqrt{a} \end{aligned}$$

Q76: If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x + y) = \frac{3}{2}$, then $\sin x + \cos y$ is equal to :

(A) $\frac{1+\sqrt{3}}{2}$

(B) $\frac{1-\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{1}{2}$

Q76: If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x + y) = \frac{3}{2}$, then $\sin x + \cos y$ is equal to :

(A) $\frac{1+\sqrt{3}}{2}$

(B) $\frac{1-\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{1}{2}$

Solution:

$$\cos x + \cos y - \cos(x + y) = \frac{3}{2}$$
$$\cos^2\left(\frac{x+y}{2}\right) - \cos\left(\frac{x+y}{2}\right) \cdot \cos\left(\frac{x-y}{2}\right) + \frac{1}{4} \cdot \cos^2\left(\frac{x-y}{2}\right) + \frac{1}{4} \sin^2\left(\frac{x-y}{2}\right) = 0$$

$$\Rightarrow \left(\cos\left(\frac{x+y}{2}\right) - \frac{1}{2} \cos\left(\frac{x-y}{2}\right) \right)^2 + \frac{1}{4} \sin^2\left(\frac{x-y}{2}\right) = 0$$

$$\Rightarrow \sin\left(\frac{x-y}{2}\right) = 0 \text{ and } \cos\left(\frac{x+y}{2}\right) = \frac{1}{2} \cos\left(\frac{x-y}{2}\right)$$

$$\Rightarrow x = y \text{ and } \cos x = \frac{1}{2} = \cos y$$

$$\therefore \sin x = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sin x + \cos y = \frac{1+\sqrt{3}}{2}$$

Q77: $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(2n-1)^2} \right]$ is equal to:

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) 1

Q77: $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(2n-1)^2} \right]$ is equal to:

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) 1

Solution:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(2n-1)^2} \right] \\ &= \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{n}{(n+r)^2} = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{n}{n^2 + 2nr + r^2} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \frac{1}{(r/n)^2 + 2(r/n) + 1} \\ &= \int_0^1 \frac{dx}{(x+1)^2} = \left[\frac{-1}{(x+1)} \right]_0^1 = \frac{1}{2} \end{aligned}$$

Q78: A plane passes through the points $A(1, 2, 3)$, $B(2, 3, 1)$ and $C(2, 4, 2)$. If O is the origin and P is $(2, -1, 1)$, then the projection of $\overrightarrow{OP}$ on this plane is of length:

(A) $\sqrt{\frac{2}{3}}$

(B) $\sqrt{\frac{2}{5}}$

(C) $\sqrt{\frac{2}{7}}$

(D) $\sqrt{\frac{2}{11}}$

Q78: A plane passes through the points $A(1, 2, 3)$, $B(2, 3, 1)$ and $C(2, 4, 2)$. If O is the origin and P is $(2, -1, 1)$, then the projection of $\overrightarrow{OP}$ on this plane is of length:

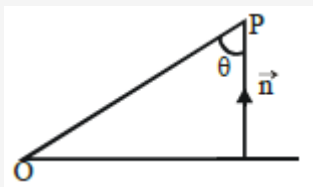
(A) $\sqrt{\frac{2}{3}}$

(B) $\sqrt{\frac{2}{5}}$

(C) $\sqrt{\frac{2}{7}}$

(D) $\sqrt{\frac{2}{11}}$

Solution:



$$\text{Normal to plane } \vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -2 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= 3\hat{i} - \hat{j} + \hat{k}$$

$$\vec{OP} = 2\hat{i} - \hat{j} + \hat{k}$$

$$\cos \theta = \frac{6+1+1}{\sqrt{6}\sqrt{11}} = \frac{8}{\sqrt{66}} \Rightarrow \sin \theta = \sqrt{\frac{2}{66}}$$

$$\therefore \text{Projection of } \vec{OP} \text{ on plane} = |\vec{OP}| \sin \theta$$

$$= \sqrt{\frac{2}{11}}$$

Q79: Let A be a 3×3 matrix with $\det(A) = 4$. Let R_i denote the i^{th} row of A . If a matrix B is obtained by performing the operation $R_2 \rightarrow 2R_2 + 5R_3$ on $2A$, then $\det(B)$ is equal to:

- (A) 80
- (B) 128
- (C) 64
- (D) 16

Q79: Let A be a 3×3 matrix with $\det(A) = 4$. Let R_i denote the i^{th} row of A . If a matrix B is obtained by performing the operation $R_2 \rightarrow 2R_2 + 5R_3$ on $2A$, then $\det(B)$ is equal to:

- (A) 80
- (B) 128
- (C) 64
- (D) 16

Solution:

$$|A| = 4$$

$$\Rightarrow |2A| = 2^3 \times 4 = 32$$

Since B is obtained by $R_2 \rightarrow 2R_2 + 5R_3$

$$\Rightarrow |B| = 2 \times 32 = 64$$

Q80: If $\alpha, \beta \in \mathbf{R}$ are such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to:

- (A) 3
- (B) 7
- (C) -7
- (D) -3

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(A) 3

(B) 7

(C) -7

(D) -3

Solution:

$\because \alpha, \beta \in \mathbf{R} \Rightarrow$ other root is $1 + 2i$

$$\alpha = -(\text{sum of roots}) = -(1 - 2i + 1 + 2i) = -2$$

$$\beta = \text{product of roots} = (1 - 2i)(1 + 2i) = 5$$

$$\therefore \alpha - \beta = -7$$

Q81: If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + x)^{2022}$ is divided by 8 is_____.

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1.00

Solution:

$$x = 4k + 3$$

$$\therefore (2020 + x)^{2022} = (2020 + 4k + 3)^{2022}$$

$$= (4(505 + k) + 3)^{2022}$$

$$= (4\lambda + 3)^{2022} = (16\lambda^2 + 24\lambda + 9)^{1011}$$

$$= (8(2\lambda^2 + 3\lambda + 1) + 1)^{1011}$$

$$= (8p + 1)^{1011}$$

$$\therefore \text{Remainder when divided by } 8 = 1$$

Q82: If the curve, $y = y(x)$ represented by the solution of the differential equation $(2xy^2 - y) dx + x dy = 0$, passes through the intersection of the lines, $2x - 3y = 1$ and $3x + 2y = 8$, then $|y(1)|$ is equal to:

Q82: If the curve, $y = y(x)$ represented by the solution of the differential equation $(2xy^2 - y) dx + x dy = 0$, passes through the intersection of the lines, $2x - 3y = 1$ and $3x + 2y = 8$, then $|y(1)|$ is equal to:

1.00

Solution:

$$(2xy^2 - y) dx + x dy = 0$$

$$2xy^2 dx - y dx + x dy = 0$$

$$2x dx = \frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right)$$

Now integrate

$$x^2 = \frac{x}{y} + c$$

Now point of intersection of lines are (2, 1)

$$4 = \frac{2}{1} + c \Rightarrow c = 2$$

$$x^2 = \frac{x}{y} + 2$$

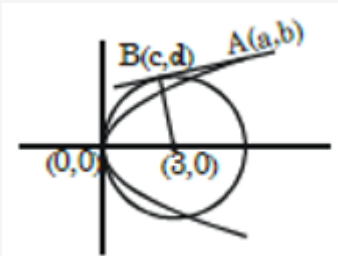
$$\text{Now } y(1) = -1$$

$$\Rightarrow |y(1)| = 1$$

Q83: A line is common tangent to the circle $(x - 3)^2 + y^2 = 9$ and $y^2 = 4x$. If the two points of contact (a, b) and (c, d) are distinct and lie in the first quadrant, then $2(a + c)$ is equal to:

Q83: A line is common tangent to the circle $(x - 3)^2 + y^2 = 9$ and $y^2 = 4x$. If the two points of contact (a, b) and (c, d) are distinct and lie in the first quadrant, then $2(a + c)$ is equal to:

Solution:



Let coordinate of point $A(t^2, 2t)$ ($\because a = 1$)

Equation of tangent at point A

$$yt = x + t^2$$

$$x - ty + t^2 = 0$$

Centre of circle $(3, 0)$

Now PD = radius

$$\left| \frac{3-0+t^2}{\sqrt{1+t^2}} \right| = 3$$

$$(3+t^2)^2 = 9(1+t^2)$$

$$9+t^4+6t^2=9+9t^2$$

$$t = 0, -\sqrt{3}, \sqrt{3}$$

So point $A(3, 2\sqrt{3})$

$$\Rightarrow a = 3, b = 2\sqrt{3}$$

(Since it lies in first quadrant)

For point B which is foot of perpendicular from centre $(3, 0)$ to the tangent $x - \sqrt{3}y + 3 = 0$

$$\frac{c-3}{1} = \frac{d-0}{-\sqrt{3}} = \frac{-(3-0+3)}{4}$$

$$\Rightarrow c = \frac{3}{2} \quad d = \frac{3\sqrt{3}}{2}$$

$$\Rightarrow 2\left(\frac{3}{2} + 3\right) = 9$$

Q84: If the curves $x = y^4$ and $xy = k$ cut at right angles, then $(4k)^6$ is equal to:

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4.00

Solution:

$$x = y^4, \quad xy = k$$

$$\text{For intersection } y^5 = k \quad \dots (i)$$

$$\text{Also } x = y^4$$

$$\Rightarrow 1 = 4y^3 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{4y^3}$$

$$\text{For } xy = k \Rightarrow x = \frac{k}{y}$$

$$\Rightarrow 1 = -\frac{k}{y^2} \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-y^2}{k}$$

$\therefore$ Curve cut orthogonally

$$\Rightarrow \frac{1}{4y^3} \times \left(\frac{-y^2}{k} \right) = -1$$

$$\Rightarrow y = \frac{1}{4k}$$

$$\therefore \text{ from (i) } y^5 = k$$

$$\Rightarrow \frac{1}{(4k)^5} = k$$

$$\Rightarrow 4 = (4k)^6$$

Q85: The total number of two digit numbers ' n ', such that $3^n + 7^n$ is a multiple of 10, is:

Q85: The total number of two digit numbers ' n ', such that $3^n + 7^n$ is a multiple of 10, is:

Solution:

For $3^n + 7^n$ to be divisible by 10

n can be any odd number

$\therefore$ Number of odd two digit numbers = 45

Q86: A function f is defined on $[-3, 3]$ as $f(x) = \begin{cases} \min\{|x|, 2 - x^2\}, & -2 \leq x \leq 2 \\ [x], & 2 < |x| \leq 3 \end{cases}$

where $[x]$ denotes the greatest integer $\leq x$. The number of points, where f is not differentiable is $(-3, 3)$ is _____.

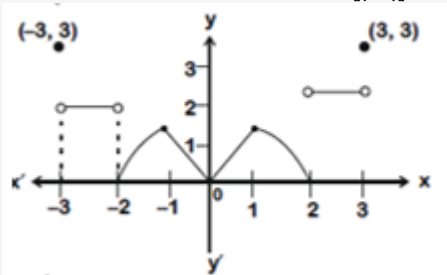
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where $[x]$ denotes the greatest integer $\leq x$. The number of points, where f is not differentiable is $(-3, 3)$ is _____.

5.00

Solution:

$$f(x) = \begin{cases} \min\{|x|, 2 - x^2\}, & -2 \leq x \leq 2 \\ [x] & , 2 < |x| \leq 3 \end{cases}$$



- ∴ The points in $(-3, 3)$ where function is not differentiable is $x = -2, -1, 0, 1$ and 2 .
 ∴ Total number of non differentiable points = 5

Q87: Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} \cdot \vec{b}$ is equal to:

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2.00

Solution:

$$\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$$

$$\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$$

$$\text{Area of parallelogram} = |\vec{a} \times \vec{b}| = 8\sqrt{3}.$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha & 3 \\ 3 & -\alpha & 1 \end{vmatrix} = \hat{i}(4\alpha) - \hat{j}(-8) + \hat{k}(-4\alpha)$$

$$\therefore |\vec{a} \times \vec{b}| = \sqrt{64 + 32\alpha^2} = 8\sqrt{3}$$

$$\Rightarrow 2 + \alpha^2 = 6 \Rightarrow \alpha^2 = 4$$

$$\therefore \vec{a} \cdot \vec{b} = 3 - \alpha^2 + 3 = 2$$

Q88: If $\lim_{x \rightarrow 0} \frac{ax - (e^{4x} - 1)}{ax(e^{4x} - 1)}$ exists and is equal to b , then the value of $a - 2b$ is:

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5.00

Solution:

$$\lim_{x \rightarrow 0} \frac{ax - (e^{4x} - 1)}{ax(e^{4x} - 1)} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{ax - (e^{4x} - 1)}{ax \cdot 4x}$$

$$\text{Use } \lim_{x \rightarrow 0} \frac{e^{4x} - 1}{4x} = 1$$

Apply L'Hospital Rule

$$= \lim_{x \rightarrow 0} \frac{a - 4e^{4x}}{8ax} \quad \left(\frac{a - 4}{0} \text{ form} \right)$$

Limit exists only when $a - 4 = 0 \Rightarrow a = 4$

$$= \lim_{x \rightarrow 0} \frac{4 - 4e^{4x}}{32x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - e^{4x}}{8x} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{-e^{4x} \cdot 4}{8} = -\frac{1}{2} \Rightarrow b = -\frac{1}{2}$$

$$a - 2b = 4 - 2\left(-\frac{1}{2}\right) = 5$$

Q89: The value of $\int_{-2}^2 |3x^2 - 3x - 6| \, dx$ is:

Q89: The value of $\int_{-2}^2 |3x^2 - 3x - 6| \, dx$ is:

19.00

Solution:

$$\begin{aligned} &= 3 \int_{-2}^2 |x^2 - x - 2| \, dx \\ &= 3 \left[\int_{-2}^{-1} (x^2 - x - 2) \, dx + \int_{-1}^2 -(x^2 - x - 2) \, dx \right] \\ &= 3 \left[\left(\frac{x^3}{3} - \frac{x^2}{2} - 2x \right) \Big|_{-2}^{-1} - \left(\frac{x^3}{3} - \frac{x^2}{2} - 2x \right) \Big|_{-1}^2 \right] \\ &= 3 \left[7 - \frac{2}{3} \right] \\ &= 19 \end{aligned}$$

Q90: A line ' l ' passing through origin is perpendicular to the lines

$$l_1 : \vec{r} = (3 + t) \hat{i} + (-1 + 2t) \hat{j} + (4 + 2t) \hat{k}$$

$$l_2 : \vec{r} = (3 + 2s) \hat{i} + (3 + 2s) \hat{j} + (2 + s) \hat{k}$$

If the co-ordinates of the point in the first octant on ' l_2 ' at a distance of $\sqrt{17}$ from the point of intersection of ' l ' and ' l_1 ' are (a, b, c) , then $18(a + b + c)$ is equal to:

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Solution:

$$l_1 : \vec{r} = (3 + t) \hat{i} + (-1 + 2t) \hat{j} + (4 + 2t) \hat{k}$$

$$l_2 : \vec{r} = (3 + 2s) \hat{i} + (3 + 2s) \hat{j} + (4 + s) \hat{k}$$

$$\text{DR of } l_1 \equiv (1, 2, 2)$$

$$\text{DR of } l_2 \equiv (2, 2, 1)$$

$$\text{DR of } l (\text{line } \perp \text{ to } l_1 \text{ and } l_2)$$

$$= (-2, 3, -2)$$

$$\therefore l : \vec{r} = -2\mu \hat{i} + 3\mu \hat{j} - 2\mu \hat{k}$$

For intersection of l and l_1

$$3 + t = -2\mu$$

$$-1 + 2t = 3\mu$$

$$4 + 2t = -2\mu$$

$$\Rightarrow t = -1 \text{ and } \mu = -1$$

$$\therefore \text{Point of intersection } P \equiv (2, -3, 2)$$

$$\text{Let point on } l_2 \text{ be } Q (3 + 2s, 3 + 2s, 2 + s)$$

$$\text{Given } PQ = \sqrt{17} \Rightarrow (PQ)^2 = 17$$

$$\Rightarrow (2s + 1)^2 + (6 + 2s)^2 + (s)^2 = 17$$

$$\Rightarrow 9s^2 + 28s + 20 = 0$$

$$\Rightarrow s = -2, -\frac{10}{9}$$

$s \neq -2$ as point lies on 1st octant.

$$\therefore a = 3 + 2 \left(-\frac{10}{9} \right) = \frac{7}{9}$$

$$b = 3 + 2 \left(-\frac{10}{9} \right) = \frac{7}{9}$$

$$c = 2 + \left(-\frac{10}{9} \right) = \frac{8}{9}$$

$$\therefore 18(a + b + c) = 18 \left(\frac{22}{9} \right) = 44$$